TARANEH SAADATI

SUMMARY

With a multidisciplinary background in mathematics, industrial engineering, energy science, and machine learning, I develop advanced forecasting models using AI to enhance the performance of predictive models. My work—published in top venues—also spans hierarchical forecasting and explainable AI, aiming to support evidence-based decision-making and contribute to the Sustainable Development Goals and a just, low-carbon future.

EDUCATION

Cardiff University - UK
2023 - 2025
Visiting PhD student - Research Assistant

Istanbul Technical University
2019 – 2025
Ph.D. in Energy Science and Technology
Thesis title: Forecasting of produced output electricity in photovoltaic power plants

The University of Kharazmi Tehran-Iran
2016 – 2018
MSc. in Industrial Engineering – Energy systems

The University of Tehran- Iran
2010 – 2015
B.Sc. in Mathematics and Applications

ACADEMIC EXPERIENCE

University of Tehran – Mathematics and Science faculty
Teacher Assistant in the field of Mathematical Logic 2014 - 2015

Winter school training in solar radiation forecasting at **Research Center O.I.E. of MINES Paris – PSL (France),** March 2023.

Projects:

- Forecasting of Produced Output Electricity in Photovoltaic Power Plants Research fellow, Istanbul Technical University, Scientific Research Projects (BAP) Project (MDK-2022-44133), Dec 2022 – Jul 2025
- Environmental Economic Dispatching problem on the standard IEEE 30-bus-6-generator test system implementing both Decomposition methods and Evolutionary methods in solving the multi objective optimization problem.
- Energy system analysis in smart houses with implementation in Tableau.
- Direct forecasting of solar irradiance time series using Artificial Neural Networks with implementation in Python, on Hatay region in Turkey as the case study.
- Feasibility study for two types of Concentrating Solar Power (CSP) technologies: Solar Tower and Parabolic Trough with implementation in System Advisor Model (SAM) software, on a field in Northern part of Egypt as the case study.
- Improving building's energy consumption by taking advantage of traditional approaches with implementation in EQUEST software, Urmia, Iran as the case study.

Conference Papers:

Nazemi. A, Saadati. T, and Yarmohammadi. M, "A review and evaluation in Energy Intensity index". Proceedings of the International Congress on Science and Engineering, 2018, Germany-Hamburg.

Nazemi. A, Saadati. T, and Yarmohammadi. M, "Multi-objective modeling of economic and environmental dispatching based on Weighted Sum approach", Proceedings of the International Congress on Science and Engineering, 2018, Germany-Hamburg.

Saadati, T., & Barutcu, B. (2024). Enhancing solar energy forecasting through Chaos theory and Echo State Networks: A case study of the Konya Eregli solar power plant. The 4th International Conference on Energy, Environment and Storage of Energy (ICEESEN). Cappadocia, Turkiye: Cappadocia University.

PERSONAL INFO

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21.05.1992 / Tehran-Iran

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Mashattan Res. Maslak,
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saadati19@itu.edu.tr
saadatit@cardiff.ac.uk

Papers Under Review:

Saadati. T, Rostami-Tabar. B, Heravi. S, (2025). Hierarchical Time Series Forecasting in National Health Service. Under review, Health Information and Libraries Journal .

Published Papers:

Saadati, T., Barutcu, B., 2024. Forecasting Solar Energy: Leveraging Artificial Intelligence and Machine Learning for Sustainable Energy Solutions. Journal of Economic Surveys. DOI: 10.1111/ joes.12678.

Saadati, T., Barutcu, B., 2024. Optimized Solar Energy Forecasting for Sustainable Development Using Machine Learning, Deep Learning, and Chaotic Models. International Journal of Energy Economics and Policy, 15(1), pp.1-11. DOI: https://doi.org/10.32479/ijeep.17766.

T. and Saadati G. Taskin, "A Sensitivity-Based Explainable Method For Remote Sensing Scene Classification," IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece, 2024, pp. 3026-3029, DOI: 10.1109/IGARSS53475.2024.10641899.

Open Science and Published Data:

Saadati, T., & Barutcu, B. (2024). Solar Energy Production Time series data, Mendeley Data, V2, Doi: 10.17632/2tpv28kr83.2.

Refrences:

- Assoc. Prof. Dr. Burak Barutcu
  barutcub@itu.edu.tr
- Prof. Dr. Bahaman Rostami-Tabar
  Rostami-TabarB@cardiff.ac.uk
- Prof. Dr. Saeed Heravi
  heravis@cardiff.ac.uk
- Prof. Dr. Gulsen Taskin
  tasking@itu.edu.tr

SKILLS

- Python
- Time Series
- Deep Learning
- Machine Learning
- Data Analysis
- Forecasting
- Artificial Neural Networks
- Ensemble Learning

Language Skills:

- Persian (Native)
- English (Fluent)
- Turkish (Fluent)

Work Experience

Presify-InavitasCo. Ankara
Data Scientist (2021 - 2022)
Analyzing Energy Consumption data, Time-series analysis, Carry out forecasting models with time-series data, Energy Management System R&D.

Scientific and Technological Research Council of Republic of Turkey (TUBITAK)
Research Scholar (Jan 2023-Mar 2024)
R&D fellow in the granted project: "A Novel Explainable AI method based on Sensitivity Analysis and its Applications to Deep Learning Models for Remote Sensing Images".

Personal Highlights

Translated Book
A child was throwing letters in to the sea (Turkish to Persian)
Author: Doğan Gündüz | Published by Berkeh | 2024